

Partical 01

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hanish@Hanish:~/2520090107_OSSP$ nano command_exec.c
hanish@Hanish:~/2520090107_OSSP$ |
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }

    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());

        execlp(command, command, NULL);

        printf("Execution failed!\n");
        exit(1);
    }

    else
    {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
hanish@Hanish:~/2520090107_OSSP$ nano command_exec.c
hanish@Hanish:~/2520090107_OSSP$ gcc command_exec.c -o command_exec
hanish@Hanish:~/2520090107_OSSP$ ./command_exec
Enter a Linux command: ls

Parent process
Parent PID: 1448

Child process
Child PID: 1453
2520090107_OSSP      command_exec.c  deadlock.c      input.txt      memory.c      mutex          producer_consumer  redirect      softlink.txt
2520090107_OSSP      command_exec.c  deadlock_prevention  lowlevelcopy  mmap.txt      mutex.c        producer_consumer.c  redirect.c    standardcopy
README.md            cow.c          deadlock_prevention.c  lowlevelcopy.c  mmapfile      output.txt     race              redirect.txt  standardcopy.c
command_exec         deadlock       hardlink.txt        memory          mmapfile.c    output2.txt    race.c            sample.txt
Child process completed.
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```

```
hanish@Hanish:~/2520090107_OSSP$ uname -a
Linux Hanish 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
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```

```
hanish@Hanish:~/2520090107_OSSP$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 20
On-line CPU(s) list:   0-19
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-13650HX
CPU family:             6
Model:                 183
Thread(s) per core:     2
Core(s) per socket:     10
Socket(s):              1
Stepping:               1
BogoMIPS:               3606.48
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant
_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
opcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx snap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize_ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    480 KiB (10 instances)
L1i:                    320 KiB (10 instances)
L2:                     12.5 MiB (10 instances)
L3:                     24 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):      0-19
Vulnerabilities:
Gather data sampling:    Not affected
Ghostwrite:              Not affected
Indirect target selection: Not affected
Itlb multihit:           Not affected
L1tf:                    Not affected
Mds:                     Not affected
Meltdown:                Not affected
Mmio stale data:         Not affected
Old microcode:           Not affected
Reg file data sampling:  Mitigation; Clear Register File
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSSB-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                   Not affected
Tsa:                     Not affected
Tsx async abort:         Not affected
Vmscape:                 Not affected
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Vmscape:                Not affected
hanish@Hanish:~/2520090107_OSSP$ lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
hanish@Hanish:~/2520090107_OSSP$ |
```

```
hanish@Hanish:~/2520090107_OSSP$ ps
PID TTY          TIME CMD
660 pts/0        00:00:00 bash
1473 pts/0        00:00:00 ps
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```

```
1473 pts/0      00:00:00 ps
hanish@Hanish:~/2520090107_OSSP$ top
top - 04:18:00 up 1:19, 1 user, load average: 0.01, 0.01, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7748.7 total, 6954.9 free, 598.5 used, 347.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7150.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21660	13144	9800	S	0.0	0.2	0:00.69	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2124	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50440	16400	15256	S	0.0	0.2	0:00.35	systemd-journal
98	root	20	0	25128	6404	5132	S	0.0	0.1	0:00.99	systemd-udev
173	systemd+	20	0	21344	13168	11048	S	0.0	0.2	0:00.12	systemd-resolve
179	systemd+	20	0	91036	8008	7044	S	0.0	0.1	0:00.20	systemd-timesyn
185	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
186	message+	20	0	9604	5416	4692	S	0.0	0.1	0:00.17	dbus-daemon
211	root	20	0	17980	8832	7792	S	0.0	0.1	0:00.11	systemd-logind
243	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
244	syslog	20	0	222516	5940	4564	S	0.0	0.1	0:00.08	rsyslogd
253	root	20	0	112748	25120	15168	S	0.0	0.3	0:00.13	unattended-upgr
372	root	20	0	6704	4660	3876	S	0.0	0.1	0:00.00	login
418	hanish	20	0	20328	11588	9456	S	0.0	0.1	0:00.10	systemd
419	hanish	20	0	21168	3576	1848	S	0.0	0.0	0:00.00	(sd-pam)
443	hanish	20	0	6080	5376	3708	S	0.0	0.1	0:00.01	bash
658	root	20	0	3188	1120	980	S	0.0	0.0	0:00.00	SessionLeader
659	root	20	0	3204	1260	1108	S	0.0	0.0	0:00.06	Relay(660)
660	hanish	20	0	6080	5368	3648	S	0.0	0.1	0:00.06	bash
1175	polkitd	20	0	308172	8188	7296	S	0.0	0.1	0:00.11	polkitd
1478	hanish	20	0	9496	5892	3556	R	0.0	0.1	0:00.00	top